Song or neighbours? Jury and public votes at the Eurovision Song Contest, 2016 to 2026

Technical report, mini-project, Introduction to Data Science, University of Helsinki

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1 The question

The Eurovision Song Contest is a yearly song competition that the European Broadcasting Union started in 1956 (Adamska and Reiss, 2025). Since 2016 between 25 and 26 countries have performed in each final. Every country taking part that year may vote in the final, including the ones knocked out in the semi-finals, which gave 36 to 43 voters per final in our data. Each voting country hands out two separate sets of points, one from a national jury and one from the public. Both follow the same ladder: 12 points to the favourite, 10 to the second and 8 down to 1 for the next eight, so ten countries score and the rest get nothing. Jury and public points are added up and the entry with the most points wins (Adamska and Reiss, 2025). No country can vote for itself.

Earlier studies, reviewed by Adamska and Reiss (2025), link the points to language, geographic proximity, shared borders and migration. We asked: of the variation in the points that country A gives to country B, how much is explained by B's entry and how much by the relationship between A and B? We cannot measure the song itself. The entry is represented by how well it did with every other country in the same vote. The relationship covers distance, a shared border or language, colonial ties and how the two countries voted for each other in earlier years.

Our target group are Eurovision fans and the national delegations. Fans want to know whether their country favours its neighbours. A delegation wants to know which part of its result depended on its entry.

The relationship turned out to matter far more to the public. Measured on contests the model had not seen, the entry alone explains 40.3 % of the variance in the public's points and the relationship alone 12.5 %. For the juries it is 24.9 and 2.8 % (Section 5 explains why the entry share is probably too high). A forecast from the country pair and search interest before the final places entries on average 2.1 places (juries) and 2.9 places (public) closer to their real rank than chance.

2 Data

One row of our table is one voter giving points to one finalist in one year, either as jury or as public. In 2024, for example, the Greek public gave Cyprus 12 points, while the public votes of all other countries gave Cyprus 0.89 points on average. The Greek jury gave Cyprus 10. Greece to Cyprus and Cyprus to Greece are different rows, because a vote has a direction. A voter gets a row for every finalist except itself, also when it gave that finalist nothing. The ten finals from 2016 to 2026 (there was no contest in 2020) give 19 774 such rows. The same finals have only 257 entries, which is too few to fit a model with a dozen inputs.

Results and every vote up to 2023 come from a published Eurovision dataset (Spijkervet, 2023), the years after that from the data file behind the results pages of a fan website (eurovisionworld.com). Distance, shared border, official language and colonial ties come from the GeoDist database of the French research centre CEPII (Mayer and Zignago, 2011). Two sources were built inside the group and live in Yauheni's repository (Zviazdou, 2026). One is Google Trends, which reports the search interest of a region as its share of all searches there, scaled so that the region with the highest share gets 100 (Google). We fetched it with the unofficial client pytrends (GeneralMills) for the 30 days up to the day before each final. Every entry was queried on its own, once on web search with artist and title and once on YouTube search with the title alone. A value therefore compares the voting country with the country that searched most for that entry. The other source labels the lyrics. The texts come from a public lyrics database (LRCLIB) and a large language model answered from fixed categories for mood, theme and repetition, at temperature 0. The script defaults to gpt-oss-120b on a Czech academic service, but which model actually ran is not recorded. The Spijkervet data is under CC BY 4.0 and CEPII under Etalab 2.0, while eurovisionworld.com states no licence (we only read pages its robots.txt allows) and the tables we made by hand are CC0.

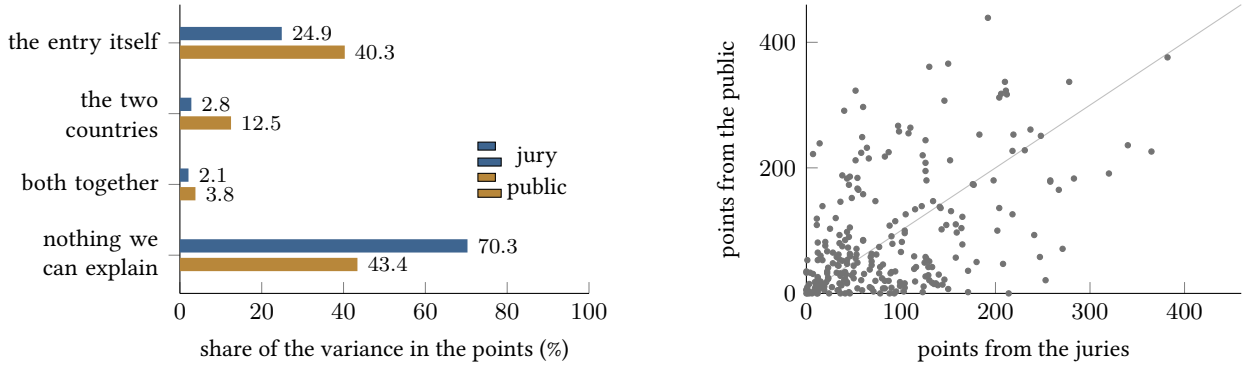


Figure 1: Left: share of the variance in the points explained on unseen finals by the entry alone, the relationship alone, their overlap and nothing (ridge regression, ten finals pooled). Right: jury against public points of the 257 entries in the finals 2016 to 2026, grey line at equal points.

The sources spell countries in three different ways. A hand made table maps all 108 labels onto three letter codes, so nothing is ever joined on a name.

Every row carries four groups of inputs. The pair group holds the logarithm of the distance, the border, language and colonial ties. It also holds how many points A gave B on average in all earlier finals since 1957 and how often A voted while B performed (before 2016 from the combined score, afterwards the mean of jury and public, both on the scale of one ladder). The attention group holds the two Trends values. The entry group holds the lyric labels and the running order, scaled from 0 for the first song to 1 for the last. The last input is the receiver effect, the average number of points that all other voters of the same kind gave the same entry. In the example above it is the 0.89.

Sixty percent of all rows are zero, because only ten of 25 or 26 finalists score. We kept the zeros, since without them the table would only contain successes. Summed per entry, jury and public points correlate with a Spearman coefficient of 0.43 over the 257 entries (Figure 1, right), so the two groups rank songs quite differently and we model them apart. Of the rows that have a Trends value, 53 % are zero on both web and YouTube, which means below the reporting threshold of Trends and not no interest. Values are missing only where nothing was queried (128 rows in the models, see Section 5).

The only personal data are the artists' names. The lyrics went to an outside language model service but are not stored in our repository. Spotify is left out because its developer terms (Spotify) forbid feeding its content into a machine learning model. YouTube view counts are left out because they include every view after the final, a bias Adamska and Reiss point out for their own model.

The pipeline is a Python package that downloads, parses, builds and fits with one command on Linux and Windows. Every raw file is checked against a stored checksum, so a rerun gives identical tables. The Trends sample and the language model answers cannot be repeated and are read from stored files. There are 85 offline tests. An export step writes the CSV files behind the figures of this report and the JSON files for the website.

3 Method

We fit two kinds of model and keep them apart. The explanatory model may use the receiver effect. The forecasting model may not, because the 0.89 points Cyprus got from everyone else are only known once everyone has voted. For the same reason it does not use the semi-final result, which is published only after the final.

Each of the ten finals is held out once while the model learns from the other nine. A random split would put votes of the same final into training and test. Missing values get the median of the training years, ridge regression picks its penalty by cross-validation inside the training years and the tree models from scikit-learn (Pedregosa et al., 2011) run with fixed settings.

The target only takes the values 0 to 12 and is zero in most rows, so a linear model is not a natural fit. We still use ridge regression for the split, because we want shares of explained variance and those add up cleanly in a linear model. The split takes three fits for each kind of vote: one with the receiver effect alone (the entry), one with the pair features alone (the relationship) and one with both. Each fit is scored with one R^2 over the predictions for all ten held out years. The part only the entry explains is R^2 of both minus R^2 of the pair fit. The part only the

Table 1: Forecast of the jury and the public ranking, gradient boosting, each final held out once. Rank error is the mean number of places between predicted and actual rank, ρ is Spearman's rank correlation, both averaged over the ten finals. Chance comes from 1000 random rankings per final, the interval is the 95 % range of the mean over ten finals.

	jury		public	
inputs	rank error	ρ	rank error	ρ
chance	8.51		8.43	
country pair	6.77	0.32	6.41	0.37
country pair + lyrics	7.50	0.21	6.78	0.29
country pair + running order	7.22	0.26	7.29	0.25
country pair + search interest	6.40	0.39	5.56	0.51
country pair + search interest + running order	7.01	0.31	6.39	0.38

chance interval: jury 7.82–9.15, public 7.75–9.03

relationship explains is R^2 of both minus R^2 of the entry fit. The overlap is the variance that either fit explains on its own but that counts only once in the joint fit. It can turn negative out of sample and we report it as it comes out. Whatever the joint fit leaves is unexplained.

As a check we repeated the split with gradient boosting, with leaves of at least 200 rows. The receiver effect leaves out the voter of the row, so a voter who gave an entry many points sees a slightly lower value. Smaller leaves would let boosting learn that artificial pattern.

For every directed pair we also computed how many points A gave B beyond what the entry model predicted for that vote, where the prediction comes from a model that had not seen that year. The mean over all finals in which A voted and B performed is what we call affinity. We only show pairs that met at least three times.

The forecast fits ridge, gradient boosting and a random forest on five sets of inputs. These are the pair features alone and the pair features with the lyric labels, with the running order, with search interest or with search interest and running order together. Summing the predicted points per entry gives a predicted ranking for every final, scored with the mean rank error (how many places an entry lands from its real place, on average) and with Spearman's rank correlation. Both the rank error and the chance baseline below follow Adamska and Reiss (2025). Chance means 1000 random rankings per final. Its 95 % interval is computed for the mean over ten finals, since the model's rank error is a mean over ten finals as well. The canvas (the project plan the course asks for) named gradient boosting as the main model on 1 October, before the first results, so Table 1 shows boosting and the other two learners are only summarised in the text.

The pair history needed a second look. When 2018 is held out, the training rows of 2019 to 2026 still carry the 2018 points inside their average of earlier finals, so we reran the split and the forecasts from pair features with and without search interest with the history rebuilt in every fold from the training years only.

4 Results

Figure 1 (left) shows the split. For the juries the entry alone explains 24.9 % of the variance in the points, the relationship alone 2.8 % and both together another 2.1 %, while 70.3 % stay unexplained. For the public the numbers are 40.3, 12.5, 3.8 and 43.4 %. The relationship explains about four and a half times as much for the public as for the juries. The share of the relationship moves from year to year, between 1.5 and 3.9 % for the juries and between 5.5 and 18.0 % for the public, but in every year it is higher for the public.

The boosting check gives 22.2 and 4.3 % for entry and relationship in the jury vote and 39.7 and 18.1 % for the public. The size of the relationship share depends on the method, but boosting finds the same gap between juries and public, with a factor of 4.2 instead of 4.5. Rebuilding the pair history from the training years only changed none of the shares by more than 0.1 percentage points.

The strongest affinities are Greece for Cyprus in the public vote with 10.5 points above what the entry earned (eight meetings), North Macedonia for Albania with 10.4 (three meetings) and, among the juries, Montenegro for Serbia with 9.8 (five meetings).

Table 1 shows the forecast. Chance gives a mean rank error of 8.51 places for the jury ranking and 8.43 for the public. Every boosting model stays below the lower end of the chance interval. The pair features alone bring the error down to 6.77 for the juries and 6.41 for the public. Adding search interest helps most, to 6.40 and 5.56.

Table 2: Changes since the canvas of September.

plan in September	what we did	why
question: did the split change between voting systems?	juries against public, 2016–2026	separate jury and public points exist only from 2016
one row per entry	one row per directed pair	257 entries are too few for a dozen inputs
users of prediction markets and contestants as target group	fans and delegations	no market data left in the project
bookmaker odds and market prices as benchmark	dropped at the team meeting of 17 September	reason not recorded
Wikipedia page views	Google Trends per voting country	Trends exists per country, so per pair
migration figures from Eurostat	dropped on 20 September	we never confirmed the codes of the table
audio features	dropped on 16 September	too difficult for the course, as Yauheni argued
artist age and sex from Wikidata	dropped on 12 October	bands have neither and it is one more input per entry
semi-final result as input	dropped	published only after the final
Trends in batches next to an anchor term	each entry queried alone	the first file was 96.7 % zeros
results pages read in a browser, tasks run with make	direct download of the data file, one Python command	failed on two machines in the group

It helps the public ranking more than the jury ranking. The lyric labels and the running order make boosting worse whenever they are added. The random forest lands within 0.4 places of boosting in every row, while ridge is clearly worse with search interest (6.92 for the juries, 6.93 for the public). Rebuilding the pair history inside each fold moved the main rank errors by at most 0.13 places.

For the combined place in the finals of 2008 to 2024, Adamska and Reiss (2025) reach 6.05 with YouTube views alone and 6.07 with all their inputs, against 8.55 for chance. Target and years differ from ours, so this is no ranking of the two studies.

5 Limits

The receiver effect measures how an entry did with the rest of the field. That includes the bloc votes of all the other voters, so the entry share is probably too high and the relationship share too low. The relationship is a mixed bag too, since past points between two countries also carry diaspora and the quality of a country's earlier songs.

With ten finals the only uncertainty we can give for the split is its spread over the years. The best forecast is also the best of five input sets scored on the same held out years, so its gain over chance is a little optimistic.

Russia has no rows after 2021 and Belarus none after 2019. Since 2023 viewers in countries without an entry vote together as one extra voter, Rest of the World (Adamska and Reiss, 2025), which has no geography and no Trends values. In 2023 our data lists its votes as a combined score, so that year it drops out of the models (26 rows). Inside the models 128 rows have no Trends values, 76 from Rest of the World in 2024 to 2026 and 52 from Bosnia and Herzegovina in 2016, which was missing from the list of countries to query. We also do not know which language model produced the lyric labels.

6 Reflection on the canvas

Our starting point is the canvas the TA commented on in mid-September. Back then we wanted to predict scores for prediction market users and the contestants. The plan had one row per entry and used bookmaker odds, Wikipedia page views and migration figures from Eurostat. The fields for exploratory analysis and visualisation were still empty, which was the TA's main comment.

Changes and why. Table 2 lists what changed. Most of it goes back to the week of 15 September, when we moved from the entry to the directed pair as the unit and then, at a team meeting on 17 September, dropped the odds and the market prices. We did not write down why we dropped them. Without any market data the old target group had nothing left in the project, so we switched to fans and delegations.

What worked. The pair level made country effects measurable at all and keeping juries and public apart produced the main finding. Search interest per voting country was the one input that clearly improved the forecast. The pipeline now runs with one command and produced the same numbers on Tim's Windows laptop as on Linux.

What did not work and why. Every input that is the same for all voters of one entry (the lyric labels and the running order) made the tree models worse. Within a training year such a value identifies the entry, so the trees learn the training entries and fail on a new year. Our canvas planned to query Trends in batches next to a fixed anchor term. The first Trends file that came back was 96.7 % zeros. It only became usable after every entry was fetched on its own. The win probabilities, simulated by adding resampled residuals to the predictions, were badly calibrated: the eight entries given more than 50 % won the public vote once, while the 15 entries given 1 to 5 % won it three times. For weeks the pipeline only ran on one laptop, until the browser based scraping failed for Yauheni and on a Windows laptop (Table 2).

How we worked. We talked in a Telegram group and met for a brainstorming session on 17 September. On 20 September we agreed on a task sheet that gave each block of data an owner. Yauheni kept his data and scripts in his own repository (Zviazdou, 2026). The main pipeline reads it at a fixed commit and our changes to it went in as a pull request. Yauheni asked whether we had tried a random forest and whether the finished tables could be downloaded as files. Both are now part of the pipeline.

Who did what. Yauheni built the first dataset of results from 2008 to 2025. He also collected the lyrics, labelled them with the language model and proposed dropping the audio features. Anna-Ilona owned the popularity and artist data, worked on the data collection part of the canvas and delivered the first Trends file. Tim wrote the pipeline and the models, fetched the second Trends file and built the website and the presentation slides.

What we would do differently. We would fix the unit and the question before choosing sources, since within a week of handing in the canvas the target group, the unit and three of the planned sources had changed. We would test every source on a small sample first and run the pipeline on every group member's machine in the first week.

Future steps. A model made for rankings would fit the target better than ridge regression and bootstrapping over years would give the split an interval. The combined scores before 2016 could show whether the role of the relationship changed over time, which was our very first question.

Visualisation and communication. All figures use percent and points rather than model terms. The website is a single static page that reads the exported JSON files. It opens with Cyprus, which got 12 points from the Greek public in all eight of its finals since 2016. The page then shows jury against public, the split and the strongest affinities. An explorer shows for any year and country who it gave points to and who gave points to it. A note says that a high affinity is not cheating, since neighbours often share a language, a music market or a diaspora.

Learning outcomes. Most of what we learned was about leakage. A random split leaks when rows belong together. We then found that even a history feature that only looks backwards can leak through the training years, although in our case the effect turned out to be tiny. A value measured after the event is no forecast, which ruled out YouTube views, the semi-final result and the receiver effect. The Trends zeros taught us to understand a source before querying all of it. From the tree models we learned that inputs which are constant for a whole entry invite memorising. A pipeline also counts as reproducible only once it runs on the machines of the whole group.

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